

Preparing to conduct a task analysis process

Usability professionals Courage, Redish and Wixon (2007) argue that task analysis is an activity based on four core principles:

1. It is an integral part of a broader analysis that includes understanding users and their environments.
2. Task analysis includes understanding users' goals.
3. Although the focus, methods, granularity, and presentation of information may differ at different times, task analysis is relevant at *all* stages of the design and development process.
4. The practical reality is that task analysis for a given project depends on many factors.

Breaking down these principles, you will notice that the first two advocate for a deep understanding of users, their environments and their goals. Since the Design Thinking methodology produces outputs during each step, it's natural to expect that task analysis is a process that has to be informed by the outputs of the previous phase—i.e., empathizing with your users. During this step, you may have already undertaken **user interviews**, or collected data through observing how your users go about their daily lives, in order to understand them better and build empathy with them. In short, you will have engaged in some **user research**, which may result in several outputs such as **user personas**, scenarios and **storyboards**. All of this data is essential for task analysis, as you will base your work according to these outputs.

Of course, just *collecting* any data during your user research is far from enough. If you plan to use task analysis (or indeed many of the other tools in a UX designer's skillset, such as customer journey maps), your **data collection** must be focused.

Larry Marine, a UX consultant, argues that your user research should focus on collecting the following five types of data, which you will use later during the task analysis phase:

- **Trigger:** What prompts users to start their task?
- **Desired Outcome:** How users will know when the task is complete?
- **Base Knowledge:** What will the users be expected to know when starting the task?
- **Required Knowledge:** What the users actually need to know in order to complete the task?
- **Artifacts:** What tools or information do the users utilize during the course of the task?

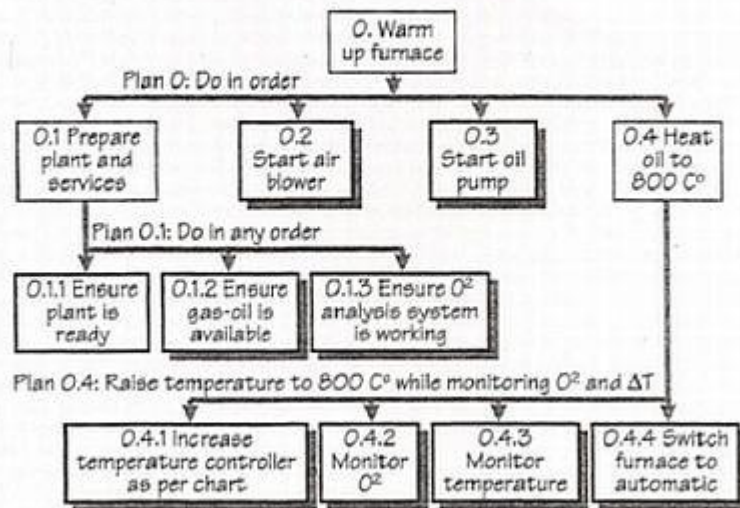
First steps in Conducting a task analysis

Armed with the information you gathered during the empathy phase, you can then begin to sketch out how a user goes about his or her daily life by mapping out the sequence of activities required to achieve a goal. Before you begin, it's important to have an overview of the process and its steps, so you can better prepare.

According to the UXPA's [Usability](#) Body of Knowledge Site, the process of task analysis can be broken down into the following steps:

- Identify the task to be analyzed: Pick a persona and scenario for your user research, and repeat the task analysis process for each one. What is that user's goal and [motivation](#) for achieving it?
- Break this goal (high-level task) down into subtasks: You should have around 4–8 subtasks after this process. If you have more, then it means that your identified goal is too high-level and possibly too abstract. As [Don Norman](#) (1998) said, users are notoriously bad at clearly articulating goals: e.g., "I want to be a good mom" – where do you even begin? Each subtask should be specified in terms of objectives. Put together, these objectives should cover the whole area of interest—i.e., help a user achieve a goal in full.
- Draw a layered task diagram of each subtask and ensure it is complete: You can use any notation you like for the diagram, since there is no real standard here. Larry Marine shares some helpful advice on the notation he uses, which is examined below.
- Write the story: A diagram is not enough. Many of the nuances, motivations and reasons behind each action are simply lost in the diagram, because all that does is to depict the actions and not the reasons behind them. Make sure you accompany your diagram with a full narrative that focuses on the *whys*.
- Validate your analysis: Once you're happy with your work, review the analysis with someone who was not involved in the decomposition, but who knows the tasks well enough to check for consistency. This person can be another team member working on the same project, but you could also enlist the help of actual users and stakeholders for this purpose.

One more trick you might wish to consider is conducting a parallel task analysis. This means simply to get more than one person on your UX team to undertake the process simultaneously—so that you can later compare your outputs and merge them into a final deliverable. This can be especially helpful if you are working internationally, or where multiple personas have to be considered for the same goal.



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An example layered diagram from a task analysis process (goal: warm up a furnace)

Types of Task Analysis

You can approach a task analysis activity from two main viewpoints. So far, we have discussed breaking down tasks into sub-tasks, an approach termed Hierarchical Task Analysis. However, you can also differentiate your viewpoint by focusing on tasks that require decision-making, problem-solving, memory, attention and judgment. This process is called a **Cognitive Task Analysis**. From this viewpoint, you would be concerned not just with how the actual activities involved in meeting a goal are performed but also with finer details that aim to uncover how a novice might perform compared to an expert, the level of **cognitive load** required for each step, how experts make decisions, or how users develop **mental models** for an activity which are later reused or adapted for other purposes.

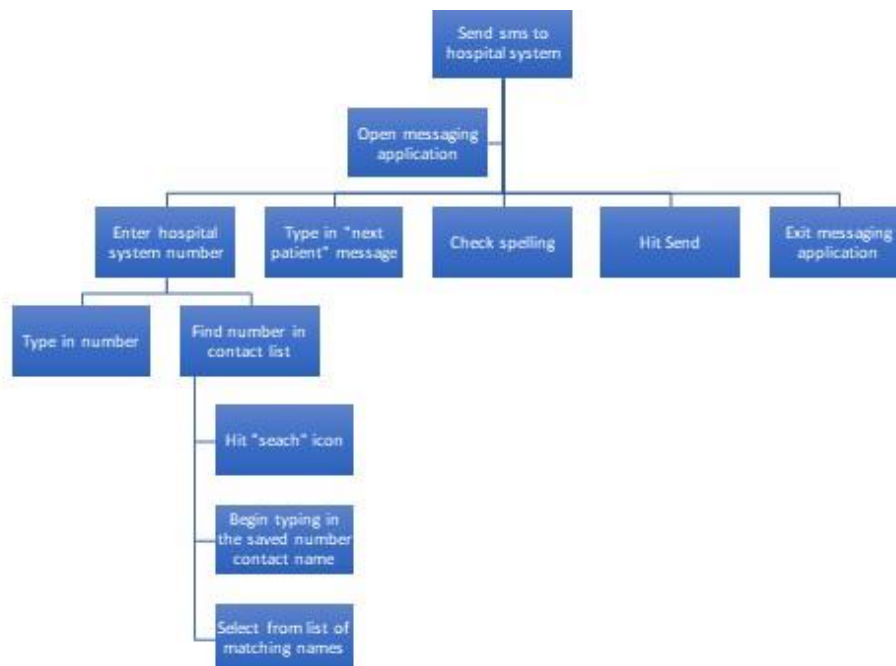
An example of Task Analysis

Let's now work through an example to illustrate how you might undertake a task analysis. Assume that Rosie, a home-visit doctor, needs to update the hospital's system with her whereabouts (i.e., that she's left one patient to go to the next) via text message. Rosie uses a custom device which appears to be a mobile phone but is tailored for use by doctors. All she needs to do is send a short SMS-like text with the words "next patient" to a certain number.

Goal: 1. Send a text message to hospital's system

Subgoals:

- 1.1. Open the messaging application on her mobile phone.
- 1.2. Enter the hospital system's special number.
- 1.3. Move to the text input field.
- 1.4. Type the words "next patient".
- 1.5. Check the spelling (because it needs to be precise for the system to accept it).
- 1.6. Hit the "send" button.
- 1.7. Exit the messaging application.



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A sample task analysis diagram for sending a short message to the hospital system. Note how each step represents a sub-task that can be broken down into further sub-tasks.

In the above diagram, you will notice that we have identified different “plans” about how Rosie might achieve a certain goal. In the diagram, pretty much everything except for displaying matching contact names is performed by the user. We can see that Rosie is not getting much support here. Perhaps we can intervene by redesigning the device’s messaging application.

- We can eliminate the “hit search icon” step by showing a list of matching contacts to Rosie as soon as she begins typing any letters or numbers in the “To:” text input field.
- We could even eliminate the typing by displaying a “frequently used contacts” dropdown under the “To:” field, even when Rosie hasn’t yet typed anything.
- We can help Rosie spellcheck her message by highlighting the words that are spelled incorrectly or by causing the device to vibrate whenever a word is misspelled.
- In order to prevent errors, we could disable the “send” button if spelling mistakes are present in the text.
- We can eliminate the “exit message” step by automatically closing the messaging application once the message has been sent.

You will note that some of the ideas for improvement discussed here are based on the simple hierarchy of tasks (i.e., eliminating needless steps), whereas others depend on a cognitive analysis approach (i.e., lessening the mental workload of the user). What we ultimately decide to do will depend on *how* the user *visualizes* the problem. If the users believe that entering correctly spelled text is the source of their greatest frustration, then our design should focus on that aspect. Solving the real issues has the biggest impact on the project, and you should continuously strive to see the task analysis from the user’s perspective and *not* what you could actually do (or want to do).

“In my 25 years of experience on over 250 projects, not one single product has been focused on solving the right problem.”

—Larry Marine, *UX consultant*

Larry Marine likes to annotate his task analysis diagrams using different colors in the various flows:

- Green represents the actions that users need to do.
- Yellow represents a step the system can do.
- Purple represents objects, tools, or information that the users need.
- Orange represents questions or issues about the task.

A task analysis would probably have a greater proportion of “green” flows originally. A re-designed task would probably have fewer “green” and more “yellow” flows, to show that you’ve really managed to off-load tasks from a user to a system, thus improving their overall experience by making their lives easier.

The Take Away

Task analysis is one of the most powerful tools in a UX designer’s skillset. As you will have undoubtedly seen, it’s not hard to get to know how to do it. However, the difficult part is remembering to keep the user’s perspective and resist the temptation to generate your own interpretations of the problem, or to “stick in” bits of design just for the sake of doing it. Remember also that task analysis is useless when it isn’t backed by rigorous user research. Without user research data, any efforts to proceed with task analysis will be in the blind and will result in

failure, because you will be capturing mostly what *you* think the problem involves, as opposed to what the *users' actual* needs are. Remember also that task analysis is not a one-off process. You can repeat it on your own designs later in the process, since Design Thinking is an iterative process that will eventually lead you back to the “define” stage at some point. Finally, remember that—like any other activity in UX design—task analysis requires time, resources, people and budget. Be sure to balance these requirements carefully and engage in the process only if you have a sufficient amount of all!

References & Where to Learn More

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